MIL-N-43158B
4 February 1977
SUPERSEDING
MIL-N-43158A
23 January 1967

MILITARY SPECIFICATION

NOODLES, CHOW MEIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for chow mein noodles for use by the Department of Defense as an item of general issue.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-T-425	-	Tinplate (Hot Dip and Electrolytic)
TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistance Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8920

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE

Regulations Governing the Grading and Inspection of Egg Products

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Poultry Division, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN OIL CHEMISTS' SOCIETY

Official and Tentative Methods of the American Oil Chemists' Society

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South Sixth Street, Champaign, IL 61820.)

ASSOCIATION OF OFFICIAL ANAYLTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards of Grades for the Dry Milk Industry, including Methods of Anaylsis; Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. All materials shall be of good quality. They shall be clean, sound, wholesome, and free from evidence of insect infestation or foreign matter.

3.1.1 Farinaceous ingredients. The farinaceous ingredients from which noodles are made shall consist of either semolina, durum wheat flour or hard wheat flour or a combination thereof. The farinaceous ingredients shall comply with the Definitions and Standards of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.1.2 Salt. Salt shall be clean, white, free-flowing, refined sodium chloride. Iodized salt shall not be used.

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3.1.3 Eggs, whole, dried. Whole dried eggs shall have been produced and handled in compliance with requirements contained in the USDA publication entitled, "Regulations Governing the Grading and Inspection of Egg Products". The whole dried eggs shall have been prepared under continuous inspection of the USDA and shall bear the inspection shield mark. Whole dried eggs shall have been laboratory tested and certified salmonella negative by the USDA.

3.1.4 Baking powder. Baking powder when used, shall be of a free-flowing, commercially available grade.

3.1.5 Vegetable oil shortening. The vegetable oil shortening shall be refined, hydrogenated or partially hydrogenated coconut oil, peanut oil, cotton seed oil or soybean oil or a combination thereof. The oil shall be thoroughly refined, deodorized and free from rancidity and objectionable flavor after heating to 400°F (204°C) and shall have a free fatty acid content of not more than 0.08 percent expressed as oleic acid and an AOM of not less than 100 hours prior to its first use. In addition, the frying oil shall be stabilized with equal amounts (0.01 percent by weight of each) of butylated hydroxyanisole and butylated hydroxytoluene allowable by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.2 Formulation. The wet noodles shall be prepared from farinaceous ingredients, water, egg products and salt. Baking powder may be used as an optional ingredient. These ingredients shall be mixed and combined to form a dough of firm texture, uniform composition, and consistency. The finished dough shall be formed into strips or ribbons of uniform cross-sectional dimensions.

3.2.1 Processing. The strips or ribbons of wet dough, formulated according to 3.2, shall be fried by complete and continuous submersion in the heated vegetable oil shortening. The oil shall be maintained at a constant temperature. The submersion and frying time shall be so controlled that the finished product is of uniform composition and color. Each day's production shall be started with a clean frier and new oil; or freshly filtered oil.

3.3 Finished product. The finished product shall be crisp, free from bitter, musty, rancid, charred or other uncharacteristic flavor or odors. It shall contain not more than 3.0 percent moisture, 2.2 ± 0.7 percent by weight of chlorides (calculated as NaCl), and shall not have more than 1 percent, by weight, of defective units, such as heat damage or discolored units, as evidenced by an orange, pink, red, brown or black discoloration. The finished product shall possess bright, uniform chow mein noodle color, approximating FED-STD-595 color chip 23594. The finished noodle shall contain not more than 10 percent, by weight, of noodles less than 1 inch (25 mm) in length and shall be 1/8 ± 1/32 inch (3.2 ± 0.8 mm) in width and thickness.

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3.4 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.5 Workmanship. The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practices. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection. The product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients. The sample unit for examination shall be 1 pound (0.45 kg) of ingredient extracted from a primary container. Five containers shall be sampled or all containers in the event a lot consists of less than five containers. Examination shall be made of labels, invoices, or other valid documents, as applicable, to determine conformance to identity and quality (kind, grade, as applicable) requirements. In addition, each ingredient shall be examined organoleptically as necessary, to determine conformance to condition requirements of 3.1 and as may be contained in the pertinent ingredient paragraph. A nonconformance shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

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4.3.1.2 Testing of components. Testing of components shall be in accordance with table I. Test requirements shall be average requirements. A test failure shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Reqmts & test ref para
Vegetable oil shortening	One pound (0.45 kg) composite 1/	Pounds (kilograms)	S-1	Stability AOM	2/	3.1.5 4.5.1
				Free fatty acid	0.01 percent expressed as oleic	3.1.5 4.5.1
Cans	One can w/lid	Cans	S-1	Tin plate	Nearest 0.01 lb/base box	4.5.3 5.1.1
				Base plate weight	Nearest lb/base box	4.5.5 5.1.1

1/ Composite shall be derived from the number of primary containers indicated by inspection level.

2/ Report as "pass" if results are acceptable and report to the nearest hour of stability if results are unacceptable.

4.3.2 In-process examination. Process examination shall be performed to determine compliance with the formulation and frying requirements in 3.2 and 3.2.1, respectively. Formulation records shall be maintained. Failure to comply with any of the requirements shall be basis for rejection of the end item.

4.3.3 Examination of end item. Examination shall be in accordance with table III and IV, 4.3.3.1 and 4.3.3.3. Sampling plans and acceptable quality levels (AQLs) for aforementioned table shall be in accordance with table II. AQLs shall be expressed as percent defective for table III and defects per hundred units for table IV.

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TABLE II. Sampling plans and AQLs

Examine table	Sample unit	Lot size expressed in	Inspection level	AQLs		
				Major A	Major B	Minor
III	One filled can	Cans	S-2	-	-	1.5
IV	Two ounces from a can	Cans	S-3	1/	1.5	6.5

1/ Finding of a Major A defect shall be cause for rejection of the lot.

TABLE III. Examination for net weight 1/ 2/

Category	Defect
Minor	
201	Net weight less than 1 pound, 7-1/2 ounces (670 g).

1/ Report results to nearest 1/4 ounce (nearest 10 g).

2/ Lot shall be rejected if sample data indicate lot average net weight less than 1 pound, 8 ounces (680 g).

TABLE IV. Examination of product

Category			Defect
Major A	Major B	Minor	
	151		Color not as specified.
	152		More than 1 percent by weight of defective units (e.g., heat damaged, discolored).
	153		More than 10 percent by weight less than 1 inch (25 mm) in length.
		201	Width or thickness less than 3/32 inch (2.4 mm) or more than 5/32 inch (4.0 mm). 1/
	154		Not crisp.
101			Undesirable flavor or odor (e.g., bitter, musty, rancid, charred).
102			Presence of foreign material (e.g., metal, glass, paint, weed, filth, insects, insect parts).

1/ Examine one noodle per sample unit.

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4.3.3.1 Examination of cans. Examination of cans for external condition, and leakage or vacuum, as applicable, shall be in accordance with the pertinent examination criteria in PPP-C-29. In addition, the following defects shall be included in the external examination of cans. Major - Nomenclature missing, incorrect or illegible; Minor - Other required label information missing, incorrect or illegible. Examination of external can coating for level A packaging shall be in accordance with the quality assurance provisions of TT-C-495.

4.3.3.2 Examination of shipping containers. Shipping containers shall be examined in accordance with the examination criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination: Major - Markings missing, incorrect or illegible; Minor - Arrangement or number of cans not as specified. For level C pack only the defects for arrangement or number of cans, and marking shall be applicable.

4.3.3.3 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.3.4 Test for oxygen content of cans. When cans are gas packed, they shall be tested for oxygen content in accordance with 4.5.2. The sample unit for testing shall be one filled and sealed can and the sample size shall be the number of cans indicated by inspection level S-2. The cans selected shall show no leakage when submitted to the leakage examination. Lot size shall be expressed in terms of cans. The acceptable quality level shall be 2.5 expressed as percent defective. Results shall be reported to nearest 0.1 percent and a defect shall be scored for a result exceeding 2.0 percent.

4.4 Test of finished product. The finished product shall be tested for moisture and salt content in accordance with 4.5.1. The sample for testing shall be a one pound (0.45 kg) composite of product derived from the number of cans indicated by inspection level S-2. Lot size shall be expressed in cans. Results shall be reported to nearest 0.1 percent. A test failure shall be cause for rejection of the lot.

4.5 Test methods. Test procedures which differ from those specified herein, unless otherwise excepted, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, test methods specified herein will govern.

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4.5.1 Test procedures.

4.5.1.1 Chemical analysis. Chemical analysis shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists and the American Oil Chemists' Society, as specified:

Test	Source	Method
Vegetable oil shortening		
Stability (AOM)	1/	Cd-12-57
Free fatty acid content (calculated as oleic)	2/	Ca-5a-40
Finished product		
Moisture	Chapter: Cereal Food Section: Macaroni, Egg Noodles and Similar Products	Total solids and moisture (vacuum oven method)
Chlorides		Chloride in ash as sodium chloride

1/ 2/ Official and Tentative Methods of the American Oil Chemists' Society.

4.5.2 Oxygen content of cans. Oxygen content in headspace of cans shall be determined in accordance with American Dry Milk Institute, Inc. Publication, Bulletin 916, Standard for Grades of Dry Milks, Chapter of Analysis: Determination of Oxygen.

4.5.3 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.5.4 Test for leakage. The filled and sealed can shall not leak when tested in accordance with PPP-C-29.

4.5.5 Base plate weights. Test for base plate weight and permissible variations in lot weight shall be as specified in QQ-T-425.

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5. PACKAGING

5.1 Preservation-packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A net weight of 1 pound, 8 ounces (680 g) of the product shall be filled into a size 603 x 700 open-top style, round, metal can. The can shall be made throughout from not less than commercial 0.25 pound per base box electrolytic tin plate (see 4.5.3) and shall be coated overall on the outside with a coating in conformance with type I or II or when specified (see 6.1), type III of TT-C-495. The product shall be gas-packaged by first removing the air from the filled can and replacing with nitrogen. If tip soldering is used to close the exhaust vent for nitrogen filling, a suitable noncorrosive flux shall be used. If the end of the can is clinched prior to vacuumizing and gassing, the flange end or lid shall be indented or dimpled to prevent seal-off trouble. The oxygen content of the gases in the hermetically sealed container, not less than 3 days after filling, shall not exceed 2.0 percent when tested in accordance with 4.5.2. Alternatively, the can shall be evacuated before sealing to obtain a gage vacuum of not less than 26 inches (not more than 13 kPa absolute). When vacuum packaged, the can body shall be beaded and shall have a base of 112 pounds per base box (see 4.5.3), and the can ends shall have a base plate weight of 100 pounds per base box (see 4.5.5). The filled and sealed can shall not leak when tested in accordance with 4.5.4.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with the level A, B, or C requirements of PPP-C-29, for size 603 by 700 cans, as specified (see 6.1).

5.3 Unitized loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Marking.

5.4.1 Labeling. Labeling of cans shall be in accordance with MIL-L-1497.

5.4.2 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unitized loads. Unitized loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) When type III can coating is required (see 5.1.1).
(d) Type and class of unit load when unitized loading is specified (see 5.3).

6.2 Can defects. For classification of can defects, the following document is suggested as a guide; Visual Inspection Gauge (Metal) Food Containers. Single copies may be obtained from the Deputy Administrator, Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.

6.3 Color chips. The standard 3 x 5 inch (76.127 mm) color chips may be obtained from the General Services Administration, Region #3, Washington, DC 20407.

6.4 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/ NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8920-0226

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-N-43158B Noodles, Chow Mein

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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